Installation Guide: ROS 2 Humble on Ubuntu 22.04 (VirtualBox / VMware)

This document provides a professional, step-by-step procedure for installing ROS 2 Humble Hawksbill on an Ubuntu 22.04 (Jammy Jellyfish) virtual machine hosted via VirtualBox or VMware.

Prerequisites

- VirtualBox or VMware installed on the host machine.
- A newly configured virtual machine running Ubuntu 22.04.
- An active internet connection to download necessary packages.

Step 1: Configure Locale (UTF-8)

Ensure that your system locale supports UTF-8. If you are using a minimal environment, verify your current locale by running the following command:

```
locale
```

If the output does not indicate UTF-8 support, execute the following commands one at a time to set it up:

```
sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8
```

Step 2: Set Up Sources and Repositories

Enable the Ubuntu Universe repository to allow access to necessary packages:

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

Install the ROS 2 apt source packages to automatically configure the ROS 2 repositories and keys for your system:

```
sudo apt update && sudo apt install curl -y
export ROS_APT_SOURCE_VERSION=$(curl -s
https://api.github.com/repos/ros-infrastructure/ros-apt-
```

```
source/releases/latest | grep -F "tag_name" | awk -F"" '{print $4}')
curl -L -o /tmp/ros2-apt-source.deb "https://github.com/ros-infrastructure/ros-apt-source/releases/download/${ROS_APT_SOURCE_VERSION}/ros2-apt-source_${ROS_APT_SOURCE_VERSION}.$(. /etc/os-release && echo ${UBUNTU_CODENAME:-${VERSION_CODENAME}})_all.deb"
sudo dpkg -i /tmp/ros2-apt-source.deb
```

Step 3: Install ROS 2 Packages

Since ROS 2 packages are frequently updated, update your apt repository caches and upgrade existing packages before installation:

```
sudo apt update
sudo apt upgrade
```

Install the Desktop version (Recommended), which includes ROS, RViz, demos, and tutorials:

```
sudo apt install ros-humble-desktop
```

Troubleshooting tip: If you encounter network errors during installation, force IPv4 and attempt the installation again using these commands:

```
sudo apt -o Acquire::ForceIPv4=true update
sudo apt update
sudo apt install ros-humble-desktop
```

Step 4: Environment Setup

To configure your environment for the current terminal session, source the setup script:

```
source /opt/ros/humble/setup.bash
```

Step 5: Automate Environment Sourcing

To avoid manually sourcing the environment every time you open a new terminal, append the source command to your bash profile.

Open your .bashrc file using a text editor:

```
gedit ~/.bashrc
```

Scroll to the very bottom of the file, paste the following line, and save the file:

```
source /opt/ros/humble/setup.bash
```

Step 6: Verify the Installation

To verify that the installation was successful, open a completely new terminal window and run:

```
ros2
```

If ROS 2 is correctly installed and sourced, the terminal will display the ROS 2 help menu along with a list of available commands.